

Отчёт по лабораторной работе №15

Настройка сетевого журналирования

Талебу Тенке Франк Устон

Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Цель лабораторной работы

Получение практических навыков работы с журналами системных событий и настройка централизованного сетевого журналирования на базе rsyslog с передачей сообщений от клиента к серверу по TCP.

Выполнение лабораторной работы

Настройка сервера сетевого журнала

```
(gedit:9739): dconf-WARNING **: 17:32:11.810: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server ~]# systemctl restart rsyslog
[root@server ~]# lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
       Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portalf file system /run/user/1000/doc
       Output information may be incomplete.
systemd      1          root    53u    IPv4          46424      0t0      T
CP *:sunrpc (LISTEN)
systemd      1          root    55u    IPv6          45670      0t0      T
CP *:sunrpc (LISTEN)
cupsd        953        root     6u    IPv6          21155      0t0      T
CP localhost:ipp (LISTEN)
cupsd        953        root     7u    IPv4          21156      0t0      T
CP localhost:ipp (LISTEN)
sshd         968        root     3u    IPv4          21197      0t0      T
CP *:down (LISTEN)
sshd         968        root     4u    IPv6          21199      0t0      T
CP *:down (LISTEN)
sshd         968        root     5u    IPv4          21201      0t0      T
CP *:ssh (LISTEN)
sshd         968        root     6u    IPv6          21203      0t0      T
CP *:ssh (LISTEN)
mariadb      1060       mysql    20u    IPv6          23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1149 mariadb    mysql    20u    IPv6          23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1215 mariadb    mysql    20u    IPv6          23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1233 mariadb    mysql    20u    IPv6          23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1238 mariadb    mysql    20u    IPv6          23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1349 mariadb    mysql    20u    IPv6          23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1352 mariadb    mysql    20u    IPv6          23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 9525 mariadb    mysql    20u    IPv6          23706      0t0      T
CP *:mysql (LISTEN)
firefox      2784       vagrant   55u    IPv4          177482     0t0      T
CP server.user.net:50258->146.75.121.91:https (ESTABLISHED)
firefox      2784       vagrant   58u    IPv4          174221     0t0      T
```

Настройка межсетевого экрана и проверка порта



Рис. 2: Проверка порта и настройка firewalld

Настройка клиента сетевого журнала

```
root@client:/etc/rsyslog.d

[vagrant@client ~]$ sudo -i
[root@client ~]# cd /etc/rsyslog.d
[root@client rsyslog.d]# touch netlog-client.conf
[root@client rsyslog.d]# gedit /etc/rsyslog.d/netlog-client.conf

(gedit:5331): dconf-WARNING **: 17:46:55.768: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

(gedit:5331): dconf-WARNING **: 17:46:55.776: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

(gedit:5331): dconf-WARNING **: 17:46:56.122: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

(gedit:5331): dconf-WARNING **: 17:46:56.129: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

(gedit:5331): dconf-WARNING **: 17:46:56.131: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

** (gedit:5331): WARNING **: 17:47:09.058: Set document metadata failed: Setting attribute metadata::gedit-spell
-language not supported

** (gedit:5331): WARNING **: 17:47:09.059: Set document metadata failed: Setting attribute metadata::gedit-encod
ing not supported

** (gedit:5331): WARNING **: 17:47:15.649: Set document metadata failed: Setting attribute metadata::gedit-spell
-language not supported

** (gedit:5331): WARNING **: 17:47:15.652: Set document metadata failed: Setting attribute metadata::gedit-encod
ing not supported

** (gedit:5331): WARNING **: 17:47:16.810: Set document metadata failed: Setting attribute metadata::gedit-posit
ion not supported

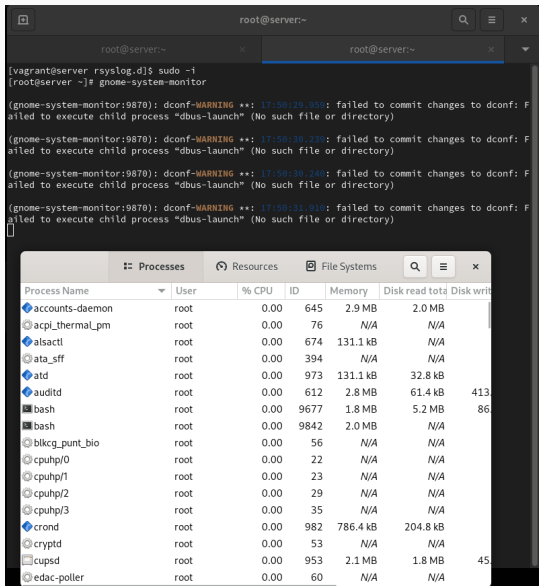
(gedit:5331): dconf-WARNING **: 17:47:16.831: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)
[root@client rsyslog.d]# systemctl restart rsyslog
[root@client rsyslog.d]#
```

Проверка приёма и отображения журналов на сервере

```
[root@server ~]# tail -f /var/log/messages
Mar 12 17:43:08 server rsyslogd[1325]: [origin software="rsyslogd" swVersion="8.2310.0-4.el9" x-p
id="1325" x-info="https://www.rsyslog.com"] exiting on signal 15.
Mar 12 17:43:08 server systemd[1]: rsyslog.service: Deactivated successfully.
Mar 12 17:43:08 server systemd[1]: Stopped System Logging Service.
Mar 12 17:43:08 server systemd[1]: rsyslog.service: Consumed 3.382s CPU time.
Mar 12 17:43:08 server systemd[1]: Starting System Logging Service...
Mar 12 17:43:08 server rsyslogd[9790]: error querying port '**UNSPECIFIED**': Servname not suppor
ted for ai_socktype [v8.2310.0-4.el9 try https://www.rsyslog.com/e/2076 ]
Mar 12 17:43:08 server rsyslogd[9790]: Could not create tcp listener, ignoring port 514S bind-add
ress **UNSPECIFIED**. [v8.2310.0-4.el9 try https://www.rsyslog.com/e/2076 ]
Mar 12 17:43:08 server systemd[1]: Started System Logging Service.
Mar 12 17:43:08 server rsyslogd[9790]: [origin software="rsyslogd" swVersion="8.2310.0-4.el9" x-p
id="9790" x-info="https://www.rsyslog.com"] start
Mar 12 17:43:08 server rsyslogd[9790]: imjournal: journal files changed, reloading... [v8.2310.0
-4.el9 try https://www.rsyslog.com/e/0 ]
█
```

Рис. 4: Просмотр системного журнала на сервере

Контроль состояния системы



The screenshot displays a terminal window with a dark theme. At the top, there are two tabs, both labeled 'root@server:~'. The terminal shows the execution of 'sudo -i' and 'gnome-system-monitor'. Below this, several log messages from 'gnome-system-monitor:9870' are visible, all indicating a 'dconf-WARNING **: 17:50:29,959: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)'. The messages are timestamped at 17:50:29,959, 17:50:30,239, 17:50:30,240, and 17:50:31,910. Below the terminal window, a 'Processes' window is open, showing a list of running processes with columns for Process Name, User, % CPU, ID, Memory, Disk read total, and Disk write total.

Process Name	User	% CPU	ID	Memory	Disk read total	Disk write total
accounts-daemon	root	0.00	645	2.9 MB	2.0 MB	
acpi_thermal_pm	root	0.00	76	N/A	N/A	
alsactl	root	0.00	674	131.1 kB	N/A	
ata_sff	root	0.00	394	N/A	N/A	
atd	root	0.00	973	131.1 kB	32.8 kB	
auditd	root	0.00	612	2.8 MB	61.4 kB	413.0 kB
bash	root	0.00	9677	1.8 MB	5.2 MB	86.0 kB
bash	root	0.00	9842	2.0 MB	N/A	
blkg_punt_bio	root	0.00	56	N/A	N/A	
cpuhp/0	root	0.00	22	N/A	N/A	
cpuhp/1	root	0.00	23	N/A	N/A	
cpuhp/2	root	0.00	29	N/A	N/A	
cpuhp/3	root	0.00	35	N/A	N/A	
crond	root	0.00	982	786.4 kB	204.8 kB	
cryptd	root	0.00	53	N/A	N/A	
cupsd	root	0.00	953	2.1 MB	1.8 MB	45.0 kB
edac-poller	root	0.00	60	N/A	N/A	

Попытка установки просмотрщика журналов

```
root@server:~  
[vagrant@server rsyslog.d]$ sudo -i  
[root@server ~]# dnf -y install lnav  
Last metadata expiration check: 1:59:28 ago on Thu 12 Mar 2026 03:52:25 PM UTC.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
---------	--------------	---------	------------	------

```
=====
```

Installing:

lnav	x86_64	0.11.1-1.el9	epel	2.4 M
------	--------	--------------	------	-------

Transaction Summary

```
=====
```

Install 1 Package

Total download size: 2.4 M
Installed size: 6.1 M
Downloading Packages:

lnav-0.11.1-1.el9.x86_64.rpm	1.8 MB/s 2.4 MB	00:01
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```
-----
```

Total

	1.1 MB/s 2.4 MB	00:02
--	-------------------	-------

```
-----
```

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction

Preparing	:	1/1
Installing	: lnav-0.11.1-1.el9.x86_64	1/1
Running scriptlet:	lnav-0.11.1-1.el9.x86_64	1/1
Verifying	: lnav-0.11.1-1.el9.x86_64	1/1

Installed:

lnav-0.11.1-1.el9.x86_64

Complete!
[root@server ~]#

Выводы по проделанной работе

Настроена система централизованного сетевого журналирования на базе rsyslog.

На сервере реализован приём сообщений по TCP, на клиенте настроена пересылка системных сообщений на сервер.

Корректность работы проверена анализом записей системных журналов.

Дополнительно выполнена автоматизация конфигурации через provisioning-скрипты Vagrant для воспроизводимого развёртывания.